

ECON 21030: Honors Econometrics

Probability Review

Spring Quarter 2024

Random variables

Much of econometrics is focused on trying to understand and characterize uncertain outcomes. Probability theory is the suite of mathematical tools used to model and analyze these ideas.

The most basic building block is a [random variable](#). Formally, a random variable is a function that assigns a number to events in the [sample space](#), which is the set of all possible outcomes of the uncertain event under scrutiny. If we were modeling the outcome of a series of n coin flips, for example, we might let X be the number of heads that come up. Often there is little harm in thinking about the sample space as the real line itself and eliding the mapping between the set of outcomes (e.g., heads, tails) and $\mathbb{R}$ that the random variable represents. That is, you can think about the experiment (i.e., the process behind the uncertain event you are modeling) as generating realizations of the random variable directly.

There are two basic types of random variables, [discrete](#) and [continuous](#). Discrete random variables can take on a countable number of distinct values. The classic example is again a coin flip, which results in either heads or tails with some probability (depending on how fair the coin is). We call the set of values X can take its [support](#) and denote it $\mathbb{X}$.

Continuous random variables take on a continuum of values. Their support may consist of the real line (i.e., $\mathbb{X} = \mathbb{R}$) or some interval (i.e., $\mathbb{X} = [0, 1]$). A good example might be tomorrow's temperature, or the time students spend reviewing material for this class. Notice that in some sense continuity is an approximation. If we model temperature with some fixed and limited degree of precision, the set of values tomorrow's weather might produce is arguably finite. Still, in many cases modeling uncertain outcome as being continuous will deliver a better (and more mathematically tractable) description of reality.

Since random variables are random, their actual, realized values are not very useful characterizations of how they behave. Telling me that a particular coin flip turned up heads, for example, is not a good description of the random process of flipping that coin. Instead, we describe random variables using their [distributions](#), which tell us the probability the random variable will take on the values in its support.

In the case of a discrete random variable, we denote its [probability mass function](#) (PMF) as $p(x)$. This function describes the probability the random variable takes value x . In the case of the coin flip, we might expect $p(0) = p(1) = 1/2$, where I am using 0 and 1 to represent the event that the result is a heads or tails, respectively.

For the PMF to be well defined, it must satisfy: $p(x) \in [0, 1]$ and $\sum_{x \in \mathbb{X}} p(x) = 1$. In words, this just means that the probability of any particular event falls between zero and one (as it must) and that the joint probability over all possible events is one. Since the joint probability for some subset $A \subseteq \mathbb{X}$ is just $Pr(x \in A) = \sum_{x \in A} p(x)$, this is equivalent to saying that $Pr(x \in \mathbb{X}) = 1$ (if that were false, $\mathbb{X}$ would not be a good description of the support of X). For convenience we sometimes also write that $p(x) = 0 \forall x \notin \mathbb{X}$, which allows us to be a little less precise about specifying the support.

In the case of a continuous random variable, we characterize its distribution using the concept of a [probability density function](#) (PDF), denoted $f(x)$. PDFs do not tell us the likelihood the continuous random variable takes the value x . Since its distribution is continuous, that probability is zero. PDFs, however, allow us to describe the probability of an event A using:

$$Pr(x \in A) = \int_{\mathbb{X}} \mathbb{1}(x \in A) f(x) dx$$

where $\mathbb{1}(x \in A)$ is the [indicator function](#) that takes value 1 if $x \in A$ and zero otherwise. As in the discrete case, we typically write that $f(x) = 0 \forall x \notin \mathbb{X}$, i.e., the density is zero outside the support. For PDFs to be well defined, they must satisfy $f(x) \geq 0$ (intuitively, negative probabilities not allowed) and $\int_{\mathbb{X}} f(x) dx = 1$ (intuitively, probabilities sum to one).

You will sometimes see people write a PMF or PDF as $p_X(x)$ or $f_X(x)$ to make it clear that are describing the distribution of the random variable X . This can be helpful when you're dealing with multiple random variables and need to be clear which you're referring to.

Finally, note that some random variables may be mixtures of discrete and continuous random variables. One often encounters these objects in econometrics when studying *censored* outcomes, e.g., a random variable that measures income but is top coded at some large value. Most of our focus in this class, however, will be on the standard discrete or continuous cases.

All random variables can also be characterized by their [cumulative distribution function](#) (CDF), denoted $F(t)$. This function describes the probability that the random variable X takes values below t : $F(t) = Pr(X \leq t)$. For discrete random variables, this is sometimes called the [cumulative mass function](#) (CMF) to emphasize the nature of the underlying random variable. As with PMFs/PDFs, they are sometimes denoted $F_X(t)$ when helpful to disambiguate between different random variables.

Since CDF/CMF describes the probability of the event $A = \{x | x \in \mathbb{X}, x \leq t\}$, in the discrete case the CMF is just $F(t) = \sum_{x \leq t} p(x)$. For continuous random variables, the CDF can be expressed as;

$$F(t) = \int_{\mathbb{X}} \mathbb{1}(x \leq t) f(x) dx = \int_{-\infty}^t f(x) dx$$

This expression also shows you that in the continuous case another definition of the PDF (owing to the fundamental theorem of calculus) is as the *derivative* of the CDF, or $F'(t) = f(t)$, whenever $f(t)$ is continuous at t .

Note that a discrete random variable's CMF is not differentiable; it "jumps" at each point in the support of X . Thus it is not true that for a discrete random variable $F'(t) = p(t)$, since this derivative is not defined at any t with $p(t) > 0$.

The CDF and CMF inherit some useful properties based on the requirements for the underlying PDFs and PMFs, specifically:

- Non-decreasing: $F(t) \leq F(t'), t \leq t'$
- Approach one: $F(\sup \mathbb{X}) = 1$
- Start at zero: $F(\inf \mathbb{X}) = 0$
- Differences: $Pr(a < X \leq b) = F(b) - F(a)$

Exercise: Prove these properties.

In a lot of what follows, we will be summing and integrating over PMFs and PDFs, respectively. Switching between summation and integration notation can be pretty tedious, so we sometimes write, for example:

$$F(t) = \int_{-\infty}^t dF(x)$$

The notation $dF(x)$ is used for the Riemann-Stieltjes integral, which allows F to either be continuous and differentiable (in which case $dF(x) = f(x)dx$) or a step function (as in the discrete case).

Expectations and moments

Cumulative distributions fully characterize random variables, but they are on their own somewhat inscrutable. It is often far more useful to use functions of these functions to describe the shape of the distribution. The [mean](#), [first moment](#) or [expectation](#) of a random variable, denoted $E[X]$, is a measure of its central tendency. In essence, the mean is just the probability weighted average value the random variable takes.

In the discrete case, the mean takes the form $E[X] = \sum_{x \in \mathbb{X}} xp(x)$. In the continuous case, it can be written as $E[X] = \int_{x \in \mathbb{X}} xf(x)dx$. Using my trick from above, we can cover both cases as $E[X] = \int_{x \in \mathbb{X}} xdF(x)$. The Greek letter μ is typically used to denote the mean of random variable X , and sometimes written as μ_X when needed for clarity.

Expectations are linear operators, meaning they satisfy the following useful properties for random variables X and Y and scalars a and b :

- $E[a + bX] = a + bE[X]$
- $E[X + Y] = E[X] + E[Y]$
- $E[a] = a$
- If $X \geq 0$, then $E[X] \geq 0$
- If $X \leq Y$, then $E[X] \leq E[Y]$

Exercise: Prove these properties using definition of the expectation for a continuous random variable.

Just as it is possible to take the expectation of X , it is also possible to take the expectation of *functions* of X . These are defined as $E[g(X)] = \int_{x \in \mathbb{X}} g(x) dF(x)$. We already encountered a version of this just above, where we considered $g(X) = a + bX$. $E[g(X)]$ is the expectation of $g(X)$, it is also the mean of the random variable defined by $Y = g(X)$. We also implicitly used an expectation earlier, since $Pr(X \in A) = E[\mathbb{1}(x \in A)]$.

One useful fact about expectations of functions of random variables is [Jensen's inequality](#), which states that $g(E[X]) \geq E[g(X)]$ for any concave function g , and the opposite for a convex function. Inequalities are strict when the functions are strictly concave (or likewise strictly convex). Jensen's inequality implies, for example, that $E[X^2] > E[X]^2$.

Higher-level moments are expectations of powers of X . The [second moment](#), for example is $E[X^2]$. Second moments are deeply connected to the [variance](#), which is the centered version: $Var(X) = E[(X - \mu)^2]$. The Greek letter σ is used to denote the square root of the variance, which is called the [standard deviation](#).

Variances measure the dispersion in the distribution of X . A simple application is [Chebyshev's inequality](#), which states that for a random variable with finite mean and variance (more on that in a bit):

$$Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

In words, this means that the probability X is greater than k standard deviations away from its mean is no more than $1/k^2$. A two standard deviation departure, therefore, can happen no more than 25% of the time.

Variances also have some simple properties, which you can verify directly as an exercise:

- $Var(X) \geq 0$
- $Var(X) = E[X^2] - E[X]^2$
- $Var(a + bX) = b^2 Var(X)$

Exercise: Prove these properties.

Different normalizations of higher-level moments are used to produce some other common descriptions of random variables. [Skewness](#), for example, is frequently defined as $E[(X - \mu)^3]/\sigma^3$ and used to measure asymmetry in distributions. [Kurtosis](#), defined as $E[(X - \mu)^4]/\sigma^4$, measures thickness of tails. Kurtosis is fun because you get to describe distributions with a lot of it as *leptokurtic*.

Moments need not exist in the sense that they need not be finite. This may seem like nonsense, but we will encounter important practical situations where you may need to deal with this phenomenon later in the course, time permitting.

Multivariate random variables

Very often we are interested in the distributions of multiple random variables, rather than just a single one. We can generalize what we've done above to consider the [joint distribution](#) of random variables X, Y using their joint CDF $F(x, y) = Pr(X \leq x, Y \leq y)$. You can think of (X, Y) as a [random vector](#), the multivariate generalization of a random variable. If both X and Y are discrete,

there will also be a joint PMF $p(x, y) = \Pr(X = x, Y = y)$. If both are continuous, there will be a joint PDF $f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$. In principle you can have mixtures of jointly distributed discrete and continuous random variables.

In some sense the multivariate case nests the marginal case just discussed above. For any random vector, we can define the [marginal distribution](#) of a component by integrating out the rest. In the discrete case, for example, we have $p(x) = \sum_{y \in \mathbb{Y}} p(x, y)$. In the continuous case we have $f(x) = \int_{y \in \mathbb{Y}} f(x, y) dy$.

We are interested in random vectors, however, because we are typically trying to understand how two uncertain outcomes are related. We may wish to know how a random American's income can be expected to relate to her height, for example, or whether companies with more employees also tend to pay higher average wages, or whether administering a particular pharmaceutical leads to lower mortality rates.

To capture notions of relatedness, we frequently work with the [conditional distribution](#) of one random variable given that the other takes a particular value. In the discrete case, conditional distributions are written:

$$\Pr(X = x | Y = y) = \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)} = \frac{p(x, y)}{p(y)}$$

This definition actually bakes in the famous [Bayes' Rule](#), which states that:

$$\Pr(X = x | Y = y) = \frac{\Pr(Y = y | X = x) \Pr(X = x)}{\Pr(Y = y)}$$

Bayes Rule is very useful because it allows you to “reverse” conditional probability statements. So much so that some people actually tattoo it on themselves. My hope is you can remember it without a tattoo, but we'll come back to this later in the course.

For continuous random variables, we work with the analogous conditional density function:

$$f(x|y) = \frac{f(x, y)}{f(y)}$$

When two random variables are not related, we say that they are [independent](#). Independence implies that the conditional distributions is the same as the marginal, e.g., $f(x|y) = f(x)$. You can see immediately that this implies that the joint mass function or density of two independent random variables is just the product of the marginal PMFs or PDFs. That is, independence implies $f(x, y) = f(x)f(y)$.

In the univariate case, it is also useful to consider moments of the conditional distributions evaluated at particular values. Of particular importance in are case is the [conditional expectation function](#), defined as $E[Y|X = x]$. This function tells us the mean of Y given that X takes the value x . In the continuous case, it is defined as:

$$E[Y|X = x] = \int_{y \in \mathbb{Y}} y f(y|x) dy$$

and analogously for the discrete case. The notation that $X = x$ is important. This is not a random variable. It is a function that evaluates to a different number for each value in the support of x . You will also see people write down the random variable $E[Y|X]$, which is just defining a new random variable $Z = g(X)$ where $g(x) = E[Y|X = x]$.

Conditional expectations have a bunch of useful properties much like regular expectations, such as that:

- $E[Y + Z|X] = E[Y|X] + E[Z|X]$
- If $Y \leq Z$, then $E[Y|X] \leq E[Z|X]$
- $E[g(X) + h(X)Y|X] = g(X) + h(X)E[Y|X]$

Conditioning is very powerful. When a random variable is in the conditioning statement, all functions of it cease to be random variables themselves. For example, $E[Yg(X)|X] = g(X)E[Y|X]$. Even though Y is a random variable, $g(X)$ is a constant once we condition on X , so the linear properties of expectations we introduced above apply.

When X and Y are independent, the conditional expectation function is very simple: $E[Y|X = x] = E[Y]$ for all x . Intuitively, because the random variables are independent, conditioning on X does not help you learn about the mean of Y . Random variables can also be [mean independent](#), a weaker condition that implies that $E[Y|X = x] = E[Y] \forall x$ but not necessarily full independence.

This type of reasoning is applied frequently through the [law of iterated expectations](#), which says that:

$$E[Y] = E[E[Y|X]]$$

This is sensible because, as we discussed above $E[Y|X]$ is itself a random variable, so it possible to take its expectation. It is straightforward to prove this result in the continuous case using some results from earlier and recalling the definitions of joint and conditional densities.

$$\begin{aligned} E[E[Y|X]] &= \int \int y f(y|x) dy f(x) dx \\ &= \int y \int f(y, x) dx dy \\ &= \int y f(y) dy = E[Y] \end{aligned}$$

We can also summarize joint distributions by taking expectations over their joint distribution, e.g., $E[g(x, y)] = \int_{y \in \mathbb{Y}} \int_{x \in \mathbb{X}} g(z, h) f(z, h) dz, dh$ in the continuous case. One of the most important characterizations of the dependence between two random variables is their [covariance](#), defined as $E[(X - E[X])(Y - E[Y])]$.

If covariance is high, then when X is above its mean Y tends to be as well. If it is negative, the opposite is true. Covariances have a handful of useful properties, which I leave to you to verify as an exercise:

- $Cov(X, Y) = Cov(Y, X)$

- $Cov(X, Y) = E[XY] - E[X]E[Y]$
- $Cov(X, a) = 0$
- $Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$
- $Cov(a + bX, c + dY) = bdCov(X, Y)$
- $Cov(X, X) = Var(X)$
- $Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2abCov(X, Y)$

Exercise: Prove these properties.

Since covariances depend on the scale of X and Y , it is also common to report the **correlation coefficient**:

$$Corr(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

The **Cauchy–Schwarz inequality** tells us that $E[XY]^2 \leq E[X^2]E[Y^2]$, which implies that $Corr(X, Y) \in [-1, 1]$ (**exercise: prove this**). You can also verify from the properties described above that correlations are *scale invariant*: multiplying either X or Y by some non-zero scalar constant will not change the correlation. This is part of the metric’s appeal.

Exercise: Use the law of iterated expectations to prove that if Y is mean independent of X , then $Corr(X, Y) = 0$.

You can also show that if Y is mean independent of X , then $E[YX] = E[Y]E[X]$.

Finally, it is sometimes useful to describe the **conditional variance**, or $Var(Y|X = x)$. This is defined much like the conditional expectation, except here we are interested in $E[Y^2|X = x] - E[Y|X = x]^2$. As in the case of conditional expectations $Var(Y|X = x)$ is a *number*, $Var(Y|X)$ is a *random variable*.

This leads to one final very useful result, the **law of total variance**. This law says that:

$$Var(Y) = E[Var(Y|X)] + Var(E[Y|X])$$

which is sometimes described as dividing the “within” X from the “between” X component of the variance of Y .

Common parametric distributions

There are handful of common distributions that are useful to known by heart. These distributions are useful because they have simple CDFs that are described by a small number of parameters. They also arise naturally in several real-world settings, as we shall see later in this course.

Continuous distributions

Perhaps the most important is the [normal distribution](#). Normals are characterized by two parameters—a mean μ and variance σ^2 , and have probability densities given by:

$$f(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

When X is normally distributed with parameters (μ, σ) , we write $X \sim N(0, \sigma^2)$. As we will see later, normals arise as the limiting distribution of many statistics and estimators, which makes them useful for quantifying uncertainty in econometric exercises. They also are “nice” in several other ways. They are symmetric and uni-modal, for example, so that the mean, median, and mode are all simply given by μ .

The normal distribution with $(\mu, \sigma) = (0, 1)$ is known as the [standard normal distribution](#). Its PDF is denoted by $\phi(x)$ and its CDF is denoted by $\Phi(x)$. It should be trivial for you to show that the density of any non-standard normal can be expressed as $f(x|\mu, \sigma) = \frac{1}{\sigma}\phi\left(\frac{x-\mu}{\sigma}\right)$.

It is also common for multiple random variables to be [jointly normal](#), in which case the joint PDF is characterized by a vector of means $\boldsymbol{\mu}$ and a variance-covariance matrix $\boldsymbol{\Sigma}$. The joint PDF for a k -vector of jointly normal random variables $\mathbf{X}$ is:

$$f(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^k |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

where $|\boldsymbol{\Sigma}|$ denotes the determinant of $\boldsymbol{\Sigma}$. It is worth familiarizing yourself with the joint normal PDF for $k = 2$ and convincing yourself of the following useful properties:

- If for jointly normal random variables X_1, X_2 $\rho = \text{Corr}(X_1, X_2) = 0$, then $f(x_1, x_2|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = f(x_1|\mu_1, \sigma_1)f(x_2|\mu_2, \sigma_2)$
- If $(X_1, X_2) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $X_1|X_2 = a \sim N\left(\mu_1 + \frac{\sigma_1}{\sigma_2}\rho(a - \mu_2), (1 - \rho^2)\sigma_1^2\right)$
- If X is normally distributed with mean and variance (μ, σ^2) then $E[X|X > a] = \mu + \sigma \frac{\phi\left(\frac{a-\mu}{\sigma}\right)}{1-\Phi\left(\frac{a-\mu}{\sigma}\right)}$, and $E[X|X < a] = \mu - \sigma \frac{\phi\left(\frac{a-\mu}{\sigma}\right)}{\Phi\left(\frac{a-\mu}{\sigma}\right)}$.
- And if X_1, X_2 are jointly normal, then $E[X_1|X_2 > a] = \mu_1 + \sigma_1\rho \frac{\phi\left(\frac{a-\mu_2}{\sigma_2}\right)}{1-\Phi\left(\frac{a-\mu_2}{\sigma_2}\right)}$, where ρ is the correlation, $E[X_1|X_2 < a] = \mu_1 - \sigma_1\rho \frac{\phi\left(\frac{a-\mu_2}{\sigma_2}\right)}{\Phi\left(\frac{a-\mu_2}{\sigma_2}\right)}$.
- If X_1, X_2 are jointly normal, then $a + bX_1 + cX_2$ is also normally distributed (exercise: derive the mean and variance of this random variable).

Exercise: Prove these properties.

Several other useful distributions build on the normal in various ways. For example, it is sometimes useful to have a random variable with support only on the positive part of the real line, which can be accomplished through the use of a [log-normal distribution](#). Log-normals are defined such that if X is log-normal with parameters (μ, σ) , then $Z = \log(X)$ is normally distributed with mean μ

and variance σ^2 . Equivalently, we might say that X is defined by the transformation $X = \exp(Z)$ applied to the normal random variable Z .

We will use squares of random variables frequently when trying to quantify uncertainty. The random variable defined by $Z = X^2$, where $X \sim N(0, \sigma^2)$ has a **Gamma distribution** with shape parameter $1/2$ and scale parameter $2\sigma^2$, denote $\Gamma(1/2, 2\sigma^2)$. Gammas scale, such that if $X \sim \Gamma(k, \theta)$, $cX \sim \Gamma(k, c\theta)$. The sum of Gammas is also a Gamma, so that if X_i is distributed $\Gamma(k_i, \theta)$ then $\sum_{i=1}^N X_i \sim \Gamma\left(\sum_{i=1}^N k_i, \theta\right)$. The mean of Γ is simply $k\theta$ and its variance is $k\theta^2$.

The **Chi-square (χ^2) distribution** is an important special case of the Gamma resulting from the square of a standard normal random variable. It is thus distributed as $\Gamma(1/2, 2)$. By the properties above, the sum of k squared standard normals is also Gamma, with distribution $\Gamma(k/2, 2)$. k is referred to as the “degrees of freedom,” and the random variable defined by this sum is said to be distributed $\chi^2(k)$. Since this is just a special case of the Γ , it has mean k and variance $2k$.

The **F distribution** (for R.A. Fisher) builds on the χ^2 and is defined by $Y = \frac{X_1/k_1}{X_2/k_2}$ where X_1 and X_2 are χ^2 with degrees of freedom k_1 and k_2 , respectively. Its CDF is typically written $F_{k_1, k_2}(x)$. F and χ^2 are related in an even more obvious sense, in the sense that $k_1 F_{k_1, \infty}$ is a $\chi^2(k_1)$ distribution.

A final related distribution is the pseudonymously named **Student’s t distribution**, which can be defined as the random variable $T = X/\sqrt{V/k}$, where X is standard normal and V is an independent $\chi^2(k)$. T is said to have a t distribution with k degrees of freedom. The t distribution arises frequently in econometrics because one can show that for a set of observations of a normal random variable, the sum of squares $\sum_{i=1}^N (X_i - \bar{X})^2$ is distributed χ_{N-1}^2 when re-scaled by the true variance σ^2 (c.f. our discussion of Gammas above, and also your first homework). Since the sum squares divided by $N - 1$ is a common unbiased estimator for variance of X , this makes t’s very useful for hypothesis testing and gives rise to “t-stats,” which we will discuss later. You can also see that t and F distributions are related in that $T^2 \sim F_{1, k}$.

Another common continuous distribution is the **uniform distribution**, which has density over support $[a, b]$ defined by:

$$f(x) = \frac{1}{b - a}$$

for $x \in [a, b]$, and zero otherwise. You often see the people working with the standard uniform density, which is just a uniform over $[0, 1]$. Uniforms are useful when you want to describe random variables that are equally likely to take on any value in their support. As an **exercise**, you can try to show that for any continuously distributed random variable X , the random variable defined by $Y = F_X(X)$ (i.e., the CDF of X applied to X) has a standard uniform distribution.

Discrete distributions

A handful of discrete distributions are also worth knowing by heart. Among the most important is the **binomial distribution**, which models the number of successes in n independent trials with success probability p . If $X \sim B(n, p)$ its pmf has support over $\{0, 1, \dots, n\}$ and is given by:

$$f(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Intuitively, this says that the probability of a sequence of n trials and k successes is $p^k(1-p)^{n-k}$. Given n trials, there are $\binom{n}{k}$ possible sequences. The binomial thus fixes n and models the likelihood of k . The [negative binomial distribution](#) does the opposite (in a sense), asking what is the distribution of number of trials required until we generate k successes at probability p . Its pmf can be written:

$$f(n; k, p) = \binom{n-1}{k-1} p^k (1-p)^{n-k}$$

Intuitively, this says that a sequence of n trials featuring k successes has joint probability $p^k(1-p)^{n-k}$. The last trial always must be a success, meaning there are $n-1$ choose $k-1$ ways the remaining trials can be arranged among the preceding trials. Note that you occasionally see the negative binomial written in terms of number of trials *before* the k th success, which will look slightly different ([exercise: how?](#)).

The [multinomial distribution](#) is the multivariate version of the binomial. In this case there are k possible outcomes (instead of two) with probabilities $p_1, \dots, p_k$, each weakly positive and total sum of one. Across n trials, we count the number of successes of each type. The resulting k -dimensional random vector $X = (x_1, \dots, x_k)$ has a multinomial distribution with pmf:

$$f(X; n, p_1, \dots, p_k) = \binom{n}{x_1, \dots, x_k} \prod_{i=1}^k p_i^{x_i}$$

where $\sum_{i=1}^k x_i = n$. Here, the product term tells us the probability of a sequence with x_i success of each type i . The term in the large brackets is the multinomial coefficient, which can be written as: $\frac{n!}{x_1! \dots x_k!}$. You can see how in the special case of $k=2$, this pmf will be identical to the binomial pmf. Note that we can also consider each element of a multinomial vector as marginally binomial. What's important is that if X is multinomial, its elements are *not* independent, since a maximum of one success of any type occur in each draw. That is, the multinomial is not the product of k independent binomials with probabilities p_k .

In the binomial / multinomial case, success probabilities are independent and constant. You might imagine this reflecting a case where we're sampling at random from a very large population, so that drawing a particular outcome k doesn't affect the probability of future k -draws. The [hypergeometric distribution](#) is a *finite* sample analogue, where we imagine there are fixed number of possible success of each type that can occur.

The classic example is that of drawing from an urn that contains a finite number of balls. Some number K of those balls may be red. The hypergeometric distribution tells us that the probability of getting k red balls when taking a n balls out of the urn at once. More generally, for n draws without replacement from population of size N that features K possible success, the probability of getting k successes is:

$$f(k; n, K, N) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

There are N choose n possible combinations of draws (i.e., samples) of size n . There are K choose k ways to get k successes. And $N - K$ choose $n - k$ ways to get $n - k$ failures. The numerator is thus the number of samples that feature exactly k success. As an **exercise**, you should show that the multinomial with $K/N \rightarrow p$ as $N \rightarrow \infty$ converges to a $B(n, p)$ distribution.

There is also a finite-sample analogue of the negative binomial known as the **negative hypergeometric distribution**, as well as a multivariate analogue.

One final distribution I will introduce is the **Poisson distribution**, which models the number of events that occur in a fixed interval when events arrive at a constant, mean rate, with the arrival of one event independent of the preceding one. For rate λ , the pfm is:

$$f(k; \lambda) = \frac{\lambda^k \exp(-\lambda)}{k!}$$

λ is the mean of the distribution, i.e., the average frequency of the event under consideration. Poissons have a few interesting features. First, both the mean and variance of the distribution of is given by λ alone. Second, its support is the set of positive integers. This is thus an example of a discrete random variable with non-finite support. Third, they are going to prove useful for a form of non-linear regression we will (hopefully) study later in this class.

Poissons arise all the time in econometrics when we are modeling the arrival rate of things like job offers. Since λ is the mean arrival rate over a fixed time interval, we naturally would get $x\lambda$ events in expectation for if we lengthen the interval by a factor of x . It is thus natural to describe the distribution of events with Poisson rate λ over a period of length Δ as rate Poisson($\lambda\Delta$). Notice that the probability at least one events in this case is given by:

$$1 - f(0; \lambda\Delta) = 1 - \exp(-\lambda\Delta)$$

The arrival rate of any offer is given by $(1 - \exp(-\lambda\Delta))/\Delta$. As we send Δ to zero, the limit of this rate is just λ . Thus it λ is sometimes described as the *instantaneous* arrival rate of the event. This is useful in models set in continuous time.

Two final, interesting, connections. First, in the $\lambda\Delta$ parameterization of the Poisson, we are describing the distribution of events in a period of length Δ with instantaneous rate λ . The distribution of times between arrival of events has an **exponential distribution** with parameter λ (**exercise: prove this.**). Exponential have support on $[0, \infty)$ and pdf $f(x; \lambda) = \lambda \cdot \exp(-\lambda x)$. The mean is given by $1/\lambda$. This connection is useful because you know that if some event is Poisson $\lambda\Delta$ over some interval Δ , you can expect to wait $1/(\lambda\Delta)$ between events.

Second, one can show that the Poisson is the limiting case of a binomial with with a very large number of trials. That is, a binomial with $n \rightarrow \infty$ and $p \rightarrow 0$ such that $np \rightarrow \lambda$ will converge to a Poisson with rate λ , i.e., $\binom{n}{k} p^k (1-p)^{n-k} \approx \frac{\lambda^k}{k!} \exp(-\lambda)$ This can be useful because for large problems the binomial pdf can be a pain to compute (**exercise: prove this.**).